Week 1 (SIT1001 Unit 1 Part 1 - 2 hours)

1.1 PROPERTIES OF PROBABILITY

In statistics, we consider experiments for which the outcome cannot be predicted with certainty. Such experiments are called **random experiments**.

Although the specific outcome of a random experiment cannot be predicted with certainty before the experiments is performed, the **collection of all possible outcomes is known** and can be described and perhaps listed.

The collection of all possible outcome (the **universal set**) of a random experiment denoted by S and is called by the **outcome space**.

Given an outcome space S, let A be a part of the collection of outcomes in S; that is, $A \subset S$. Then A is called an **event**.

In studying probability, the words set and event are interchangeable and hence the reader might want to review **algebra of sets**. Here we look at some terminologies:

- $\emptyset$ denotes the **null** or **empty** set;
- $A \subset B$ means A is a **subset** of B;
- $A \cup B$ is the **union** of A and B;
- $A \cap B$ is the **intersection** of A and B;
- A' is the **complement** of A (i.e., all element in S that are not in A). Other symbol used: A^T.

Some of these sets are depicted by the shaded regions in Figure 1.1-1, in which S is the interior of the rectangaless. Such figures are called **Venn diagrams**.

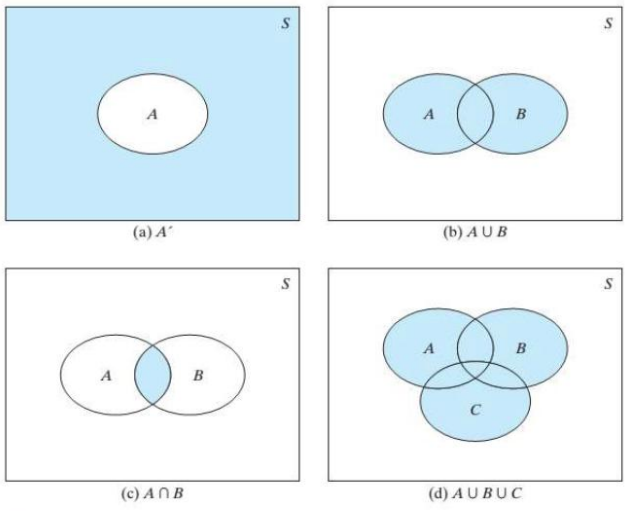


Figure 1.1-1 Algebra of sets

Some definition:

1. $A_1, A_2, \ldots, A_k$ are **mutually exclusive events** means that $A_i \cap A_j = \emptyset, i \neq j$, that is $A_1, A_2, \ldots, A_k$ are disjoint sets;
2. $A_1, A_2, \ldots, A_k$ are **exhaustive event** means that $A_1 \cup A_2 \cup \cdots \cup A_k = S$.

So if $A_1, A_2, \ldots, A_k$ are **mutually exclusive and exhaustive** events, we know that that

$$A_i \cap A_j = \emptyset, i \neq j, \text{ and } A_1 \cup A_2 \cup \cdots \cup A_k = S.$$

Set operations satisfy several properties. For example, if A, B and C are subsets to S, then we have the following:

Commutative law	$A \cup B = B \cup A$	$A \cap B = B \cap A$
Associative law	$(A \cup B) \cup C = A \cup (B \cup C)$	$(A \cap B) \cap C = A \cap (B \cap C)$
Distributive law	$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	
De Morgan's law	$(A \cup B)^T = A^T \cap B^T$	$(A \cap B)^T = A^T \cup B^T$

A Venn diagram can be used to justify the De Morgan's law:

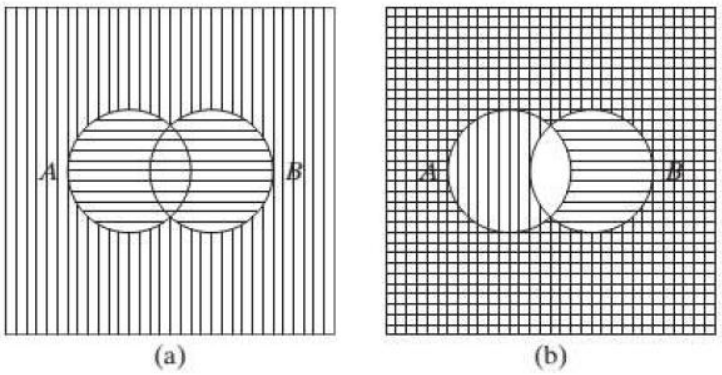


Figure 1.1-2 Venn diagrams illustrating De Morgan's laws

We are interested in defining what is meant by the probability of event A, denoted by $P(A)$.

Consider:

- Repeating the experiment a number of times, say n times.
- Count the number of times event A, say $N(A)$, actually occurred throughout these n experiments.
- The ratio $p = \frac{N(A)}{n}$ is the relative frequency of event A in these n repetition of the experiment.
- The relative frequency is usually very unstable for small values of n, but it tends to stabilize as n increases.
- Thus although we cannot predict the outcome of a random experiment with certainty, for large n, we can predict p fairly accurately.
- The number ***p* assigned to event A** is called the **probability of event *A*** and is denoted as $P(A)$.

Example 1.1

Toss a fair six-sided die.

The experiment is called a success if the face number 3 is the outcome of the roll.

Sample Space is S: {Success, Failure}.

Let A = {Sucess). We would like to assign a value to $P(A)$.

For large n, $P(A)$ will stabilize towards 1/6.

We wish to associate an event A **with** a number $P(A)$ about which the relative frequency $\frac{N(A)}{n}$ tends to stabilize with large n.

A function such as $P(A)$ that is evaluated for a set A is called a **set function**.

DEFINITION 1.1

Probability is a real-valued set function P that assigns, to each event A in the sample space S, a number $P(A)$, called the probability of the event A, such that the following properties are satisfied;

(a) $P(A) \geq 0$,
(b) $P(S) = 1$,V
(c) If A_1,A_2, A_3,... are events and $A_i \cap A_j = \emptyset, i \neq j$, then

$$P(A_1 \cup A_2 \cup \cdots \cup A_k) = P(A_1) + P(A_2) + \cdots + P(A_k)$$

for each positive integer k, and

$$P(A_1 \cup A_2 \cup A_3 \cup \cdots) = P(A_1) + P(A_2) + P(A_3) + \cdots$$

for an infinite, but countable, number of events.

The theorems that follow give some important properties of the probability set function.

THEOREM 1.1 (Theorem 1.2-1, Hogg and Tanis)

For each event A, $P(A) = 1 - P(A')$.

Proof: $S = A \cup A'$

THEOREM 1.2 (Theorem 1.2-3, HT)

If events A and B are such that $A \subset B$, then $P(A) \leq P(B)$.

Proof: $B = A \cup (A' \cap B)$

THEOREM 1.3 (Theorem 1.2-5, HT)

If A and B are two events, then

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

Proof: $A \cup B = A \cup (A' \cap B)$ and $B = (A \cap B) \cup (A' \cap B)$

THEOREM 1.4 (Theorem 1.2-6, HT)

If A, B, and C are any three events, then

$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$.

Proof:

Let a probability set function be defined on a sample space $S = \{e_1, e_2, \dots, e_m\}$, where each e_i is a possible outcome of the experiment.

If each of these outcomes has the same probability of occurring, we say that the *m* outcomes are equally likely, that is,

$$P(\{e_i\}) = \frac{1}{m}, \quad i = 1,2, \dots, m.$$

EXAMPLE 1.2

Let a card be drawn at random from an ordinary deck of 52 playing card.

a. State the sample space S.

b. State the probability of selecting one card from the deck.

c. If A is the set of outcomes that are kings, then argue that $P(A) = 1/13$.

EXAMPLE 1.3

A fair coin is tossed four times, and the sequenct of heads and tails is observed.

a. List each of the 16 sequences in the sample space *S*.

b. Let events A, B, C and D be given by

A = {at least 3 heads}; B={at most 2 heads};

C={heads on the third toss}; D={1 head and 3 tails}.

If the probability set function assigns 1/16 to each outcome in the sample space, find

$P(A)$= ; $P(A \cap B)$ = ; $P(B)$= ; $P(A \cap C)$=

$P(D)$= ; $P(A \cup C)$= ; $P(B \cap D)$=

1.2 METHOD OF ENUMERATION

EXAMPLE 1.4

Ahmad is redecorating his house and has **THREE** guest bedrooms to wallpaper.

He has **FOUR** patterns of wallpaper from which to choose.

Once a particular wallpaper has been selected for a room, that same pattern will be used for the entire room.

In how many different ways that can Ahmad wallpaper these **THREE** rooms?

Group 1:

Group 2:

Group 3:

Here, we intend to develop counting techniques that are useful in determining the number of outcomes associated with the events of certain random experiments**.**

Multiplication Principle
Suppose that an experiment (or procedure) E_1 has n_1 outcomes and, for each of these possible outcomes, an experiment (procedure) E_2 has n_2 possible outcomes. Then, the composite experiment (procedure) E_1E_2 that consists of performing first E_1 and then E_2 has n_1n_2 possible outcomes.

Permutation of *n* different objects.

Suppose that n positions are to be filled with n different objects. Thera are n choices for filling the first position, n-1 for the second, ... and 1 choice for the last position. Hence, by the multiplication principle, there are

$$n(n-1)\cdots(n-r+1)(n-r)\cdots(3)(2)(1) = n!$$

possible arrangements.

Each of the $n!$ arrangements (in a row) of n different objects is called a **permutation** of the n objects.

Permutation of *n* different objects taken *r* at a time.

If only r positions are to be filled with objects selected from n **different/distinct objects,** $r \leq n$, then the number of possible ordered arrangements is

$$n(n-1)(n-2)\cdots(n-r+1) \qquad (1)$$

That is, there are n ways to fill the first position, $(n-1)$ ways to fill the second, and so on, until there are $[n-(r-1)] = (n-r+1)$ ways to fill the rth position.

In terms of factorials, we have

$${}_nP_r \text{ or } nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n(n-1)\cdots(n-r+1)(n-r)\cdots(3)(2)(1)}{(n-r)\cdots(3)(2)(1)}$$

$$= \frac{n!}{(n-r)!}$$

Combination – unordered sample of size *r*

Each of the ${}_nC_r$ **unordered subsets** (the order of selection is not important) is called a **combination** of n objects taken r at a time, where

$$ {}_nC_r \text{ or } nC_r = (n\ r) = \frac{n!}{r!(n-r)!} $$

Ordered sample of size *r* taken from a set of *n* objects – sampling with/without replacement

If r objects are selected from a set of n objects, and if the order of selection is noted (or important), then the selected set of r objects is called an **ordered sample of size** r.

Sampling with replacement occurs when an object is selected and then replaced before the next object is selected.

Sampling without replacement occurs when an object is not replaced after it has been selected.

Hence, by multiplication principle,

a. the number of possible **ordered samples** of size r taken from a set of n objects **with replacement** is n^r.
b. the number of possible **ordered samples** of size r taken from a set of n objects **without replacement** is ${}_nP_r$.

Distinguishable permutation with two types.

Now suppose that a set contains *n* objects; n_1= *r* of one type and n_2= *n-r* of the other type.

The number of permutations of *n* diferent objects is $n!$. However, here the objects are **not all distinguishable**.

Hence, to count the number of distinguishable arrangement,

Fir select *r* out of the *n* positions for the objects of the first type. This can be done in $(n\ r)$ ways.

Then, fill the remaining positions with objects of the second type, that is $(n-r\ n-r)$=1.

Then, using multiplication principle, the number of distinguishable arrangement is

$$ (n\ r)(n-r\ n-r) = (n\ r) = \frac{n!}{r!(n-r)!} = \frac{n!}{n_1!n_2!}. $$

Distinguishable permutation with more than two types.

Suppose that in a set of *n* objects; $n_1$are similar, n_2 are similar, ..., n_s are similar, where $n_1+n_2+\cdots+n_s=n$. Then, the number of distinguishable permutation of n objects is

$$ (n\ n_1, n_2, \cdots, n_s) = \frac{n!}{n_1!n_2!\cdots n_s!} $$

EXAMPLE 1.5

A certain food service gives the following choices for dinner:

E_1: soup or tomato juice;
E_2: steak or shrimp;
E_3: French fried potatoes, mashed potatoes, or a baked potato;
E_4: corn or peas;
E_5; jello, tossed salad, cottage cheese, or coleslaw;
E_6, cake, cookies, pudding, brownie, vanilla ice cream, chocolate ice cream, or orange sherbet;
E_7, coffee, tea, milk, or punch.

How many different dinner selections are possible if one of the listed choices is made for each of E_1, E_2, ... and E_7?

■

EXAMPLE 1.6

We have four letters a, b, c, and d.

(a) List out the arrangement of the letters without repetition.

(b) State the number of permutation of the letters.

(c) State the number of possible four-letter code words using the four letters a, b, c, and d such that the letters may be repeated.

■

EXAMPLE 1.7

Determine the number of possible four letter code words, selecting from the 26 letters in the alphabet, in which all four letters are different.

■

EXAMPLE 1.8

Find the number of ways of selecting a president, a vice president, a secretary, and a treasurrer in a club consisting of 10 persons.

■

EXAMPLE 1.9

Find the number of ordered samples of 5 cards that can be drawn without replacement from a standard deck of 52 playing cards.

■

EXAMPLE 1.10

Find the number of possible 5-card hands (in 5-card poker) drawn from a deck of 52 playing cards.

■

EXAMPLE 1.11

Find the number of possible 13-card hands (in bridge) that can be selected from a deck of 52 playing cards.

■

EXAMPLE 1.12

Assume that each of the $(52\ 5) = 2{,}598{,}960$ five-card hands drawn from a deck of 52 playing cards has the same probability of being selected.

(a) Find the probability of an all-spade 5-card hand.

(b) Find the probability an event B such that exactly 3 cards are kings and exactly 2 cards are queens.

(c) Find the probability of an event C such that the set of outcomes are exactly two kings, two queens, and one jack.

EXAMPLE 1.10

A coin is flipped 10 times and the sequence of heads and tail is observed. Determine the number of possible 10-tuplets that result in four heads and six tails.

■

EXAMPLE 1.11

In an orchid show, seven orchids are to be placed along one side of the greenhouse.

(a) If there are four lavender orchids and three white orchids, determine the number of lineup of the orchids.

(b) If the colours of the seven orchids are white, lavender, yellow, mauve, crimson, orange, and pink, determine the number of different displays.

■

REMARK One can also count the number of possible samples of size r that can be selected out of n objects when the order is irrelevant and when sampling with replacement by using

$$_{n-1+r}C_r = \frac{(n-1+r)!}{r!(n-1)!}.$$

For example, if a six-sided die is rolled 10 times (or 10 six-sided dice are rolled once), how many possible unordered outcomes are there?